MILITARY SPECIFICATION

DELAY ELEMENT, DETONATING, MK 19 MOD 0

This specification is approved for use by the Naval Sea Systems Command (OS), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the minimum requirements for the fabrication, preparation for delivery, and acceptance of the Delay Element, Detonating MK 19 Mod 0, hereinafter referred to as the delay (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for

Beneficial comments (recommendations, additions deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station (611), Indian Head, Md. 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1375

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STANDARDS

Military

MIL-STD-1167	Ammunition Data Card

DRAWINGS

Naval Sea Systems Command (Code Ident 10001)

DL 1332054	Package, Shipping and Storage (For 200 Delay Element, Detonating, Mk 19 Mod 0)

PUBLICATIONS

Naval Sea Systems Command (Code Ident 10001)

WR-43	Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Item Definition. This prime item is the Mk 19 Mod 0 Detonating Delay Element. The delay is a 52 millisecond delay consisting of an aluminum body containing a delay composition, plastic primacord secured to each end of the aluminum body and sealed with an aluminum cup crimped on each end.

3.2 Classification of requirements. The requirements for the delay are classified herein as follows:

REQUIREMENTS	PARAGRAPH
First Article Sample	3.3
Design and Construction	3.4
Performance and Product Characteristics	3.5
Workmanship	3.6

3.3 First article. Unless otherwise specified in the contract or order (see 6.2), a first article sample of delays (see 4.3) shall be furnished by the contractor. The delays, their components and all materials shall be of a single type, grade, class, size, and composition and shall comply with all applicable provisions of this document. The sample shall be manufactured using the proposed full production methods.

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3.4 Design and construction.

3.4.1 Prime item documentation. This delay shall be manufactured, examined, tested and packed in accordance with contract documents, Figure 1, and this specification.

3.4.2 Classification of characteristics. Characteristics to be verified by the inspections and tests specified herein are classified as Critical, Major and Minor in accordance with WR-43. Critical characteristics are identified by the symbol (C), the Major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristic. Characteristics that are not annotated by the classification code symbol are classified as Minor.

3.5 Performance and product characteristics.

3.5.1 Performance.

3.5.1.1 Water pressure (M101). The delay shall show no signs of water leakage after being subjected to hydrostatic pressure, when tested in accordance with 4.5.1.1.

3.5.1.2 Static functioning (M102). The delay shall function producing a delay time of 52 ± 15 milliseconds, when tested in accordance with 4.5.1.2.

3.5.2 Durability and environmental.

3.5.2.1 Forty (40) foot drop (C1). The delay shall withstand 40 foot drop without burning or exploding, when tested in accordance with 4.5.2.1.

3.5.2.2 Transportation vibration (C2). The delay shall withstand transportation vibration without burning or exploding, when tested in accordance with 4.5.2.2.

3.5.2.3 Temperature and humidity (C3). The delay, packed in a hermetically sealed container, in accordance with Section 5 shall withstand temperature and humidity without burning or exploding, when tested in accordance with 4.5.2.3.

3.5.2.4 High temperature. The delay shall be conditioned at high temperature when tested in accordance with 4.5.2.4.

3.5.2.5 Low temperature. The delay shall be conditioned at low temperature when tested in accordance with 4.5.2.5.

3.6 Workmanship. The product, in all stages of production, shall be processed under such inspection control as to assure uniformity in quality. All components shall be free of chips, dirt, grit, or other foreign material. The cleaning methods used shall not damage any assembly or parts nor shall the assembly or parts be contaminated by the cleaning agents employed. There shall be no chemical or electro-chemical corrosive or deteriorative effects from the manufacturing process.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilites suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection requirements. The inspection requirements are specified by the classified characteristics in the specification and on the referenced drawings, and also by the supplemental quality assurance provisions of the contract.

4.1.2 Inspection review. When specified in the contract or order, the contractor and all test activities identified in the contract or order shall submit to the Technical Activity and Design Agent for review, a description of the proposed procedure and equipment to be used to perform all examinations and tests (see 6.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3)

2. Quality conformance inspection (see 4.4)

4.3 First article inspection. The contractor shall deliver a sample of 570 delays to the destination specified (see 6.2), for examination, testing, evaluation, and acceptance as specified in Table I. Any production of the item prior to authorization by the procuring agency shall be at the risk of the contractor. The sample shall be prepared for delivery as specified in Section 5.

4.4 Quality conformance inspection. An inspection lot shall consist of 501 to 35,000 delays offered for delivery at one time, including test samples (see 6.2). Inspection lots shall be inspected as follows:

a. Any change in materials, suppliers or production methods shall be considered an interruption for the beginning of a new inspection lot. Record reason for interruption on the Data Card (see 6.2).

b. Each lot shall be sampled as required for Plan A, Table I.

c. Upon rejection of any lot for test requirements, the next lot shall be tested and accepted in accordance with Plan A, Table I.

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d. No lot shall be accepted in accordance with Plan B, Table I, unless the preceding lot has met the test requirements of the applicable plan.

e. Sampling, examination, testing and acceptance of inspection lots shall be performed as specified in the steps given below.

STEP 1. Upon starting production or delivery of an acceptable first article sample, as applicable, all inspection lots shall be examined, tested in sequence and accepted in conformance with Plan A, Table I, until two (2) consecutive lots have been accepted.

STEP 2. Inspection lots other than those defined in Step 1 shall be examined, tested in sequence and accepted in conformance with Plan B, Table I, except that one lot shall be randomly selected from each five consecutive lots and examined, tested in sequence and accepted in conformance with Plan A, Table I.

4.5 Tests. Random samples shall be selected, tested, and accepted as specified in 4.3 or 4.4, as applicable.

4.5.1 Performance Tests.

4.5.1.1 Water pressure test. Delays shall be water pressure tested by submerging in water at a temperature of 40° F to 70° F, under a hydrostatic pressure of 43 ± 3 psig for a minimum period of three hours. Defectives are delays failing to meet the requirements of 3.5.1.1.

4.5.1.2 Static Function Test. The delay shall be placed in a partitioned steel box and tested in a shielded position to provide adequate protection for personnel and equipment. An 0.022 inch diameter insulated copper wire shall be attached to both ends of delay, by wrapping around primacord portion, for start and stop delay timing circuit. Attachment of equipment should be arranged in such a manner as to prevent premature stop of recording. The delay shall be initiated by means of an electric blasting cap and delay time shall be recorded using a chronograph capable of recording with an accuracy of ± 1/2 millisecond. Blasting caps to be used are:

Cap, Blasting, Special, Electric, J2 or M6

NSN 1375-00-028-5224-M130 (J2)
NSN 1375-00-028-5225-M130 (J2)
NSN 1375-00-756-1865-M130 (M6)

Defectives are delays failing to meet the requirements of 3.5.1.2.

4.5.2 Durability and environmental tests.

4.5.2.1 Forty (40) foot drop tests. The delays shall be subjected to the forty foot drop test prescribed in Test 103 of MIL-STD-331, except the delays shall be dropped free-fall without guidance system or associated equipment. Defectives are delays failing to meet the requirements of 3.5.2.1.

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TABLE I
INSPECTION PLANS

EXAMINATIONS AND TESTS	FIRST ARTICLE SAMPLING PLANS		QUALITY CONFORMANCE INSPECTION			
1 2 3 4 Test Sequence	Sample Size	Acceptance Criteria[1]	Plan[2] A	Plan[2] B	Sample Size	Acceptance Criteria[1]
Examination and Workmanship (4.6) (3.6)	570 Delays	Critical Ac 0 Re 1 Major Ac 10 Re 11 Minor Ac 21 Re 22	X	X	See Footnote 3	See Footnote 3
Drop Test (4.5.2.1)	10 Delays	Ac 0 Re 1	X		10 Delays	Ac 0 Re 1
Transportation Vibration Test (4.5.2.2) Water Pressure Test (4.5.1.1) Static Functioning Test (4.5.1.2)	110 Delays	Ac 0 Re 1 Ac 2 Re 3 Ac 2 Re 3	X		MIL-STD-105 Level I	Ac 0 Re 1 AQL 0.65 AQL 0.65
Temperature and Humidity Test (4.5.2.3) Water Pressure Test (4.5.1.1) Static Functioning Test (4.5.1.2)	110 Delays	Ac 0 Re 1 Ac 2 Re 3 Ac 2 Re 3			None	None None None
High Temperature Test (4.5.2.4) Static Functioning Test (4.5.1.2)	110 Delays	None Ac 2 Re 3			None	None None
Low Temperature Test (4.5.2.5) Static Functioning Test (4.5.1.2)	110 Delays	None Ac 2 Re 3			None	None None
Water Pressure Test (4.5.1.1) Static Functioning Test (4.5.1.2)	110 Delays	Ac 2 Re 3 Ac 2 Re 3	X	X	MIL-STD-105 Level I	Ac 2 Re 3 AQL 0.65
Group Data Acceptance[4] (4.5.1.2)	550 Delays	Ac 9 Re 10			None	None

NOTES:
1 Ac = Acceptance number for defectives of specified requirements of Section 3.
Re = Rejection number for defectives of specified requirements of Section 3.
AQL = Acceptance Quality Levels are specified as percent defective.
2 Test plans applicable to Inspection Lots. (see 4.4)
3 Quality assurance inspection requirements of the contract shall apply.
4 Grouped for Static Functioning Test.

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4.5.2.2 Transportation vibration test. The delay shall be subjected to the transportation vibration test prescribed in Test 104, Procedure II, MIL-STD-331. Defectives are delays that fail to meet the requirements of 3.5.2.2.

4.5.2.3 Temperature and humidity test. The delay, packed in a hermetically sealed container, shall be subjected to one 14-day temperature and humidity cycle as prescribed in Test 105, MIL-STD-331. Defectives are delays failing to meet the requirements of 3.5.2.3.

4.5.2.4 High temperature test. The delay shall be conditioned by being subjected to a temperature of 160° F ± 5° F for a minimum period of 16 hours. After conditioning, the delays shall be static functioned while at that temperature.

4.5.2.5 Low temperature test. The delay shall be conditioned by being subjected to a temperature of -65° F ± 5° F for a minimum period of 16 hours. After conditioning the delays shall be static functioned while at that temperature.

4.6 Examinations.

4.6.1 Delay examinations. Examinations shall be made to assure that the delays meet the requirements of Figure 1 and as follows. Only delays containing no defects shall be used for further testing or returned to the lot.

CLASSIFICATION OF DEFECTS

Critical

None

Major

101 Plastic primacord damaged.
102 Length from body to end not as specified.
103 Crimps not present 360° (4).

Minor

201 Length overall not as specified.
202 Body length not as specified.
203 Length of aluminum end caps not as specified.
204 Delay body diameter not as specified.

4.6.2 Packaging, packing, and marking examinations. The packaging, packing, and marking shall conform to Section 5 of this specification.

5. PACKAGING (see 4.6.2)

5.1 Packaging, packing and marking.

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5.1.1 Level A (C4).

5.1.1.1 Two hundred (200) delays shall be packaged, packed, and marked in accordance with DL 1332054.

5.2 Ammunition data card. Data cards shall be prepared in accordance with MIL-STD-1167 (see 6.2).

6. NOTES

6.1 Intended use. The Delay Element, Detonating, Mk 19 Mod 0 is a short interval delay device used to produce a 52 millisecond delay between demolition charges.

6.2 Ordering data. DD Form 1423 and other related procurement documents should specify:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Mark and mod being ordered.

c. Quantity being ordered.

d. First article sample, if applicable, and test facility where sample is to be shipped for evaluation.

e. Weapons Quality Engineering Center coordinating surveillance.

f. Inspection lot size if different than 4.4.

g. Inspection review when applicable (4.1.2).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations (ASPR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

PARAGRAPH	DATA REQUIREMENT	APPLICABLE DID
4.4	Quality Conformance Inspection	*UDI-E-20602
5.2	Ammunition Data Card	DI-E-2001

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*Exceptions, preprinted appendant test report forms will not be furnished with this specification. Test reports with individual data will be forwarded to the Technical Activity and Design Agent.

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 Conflicting requirements. Refer in writing to the procuring agency for clarification, any conflicting requirements arising between this specification and any specification or publication listed herein.

6.4 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Armed Services Procurement Regulations (ASPR) 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

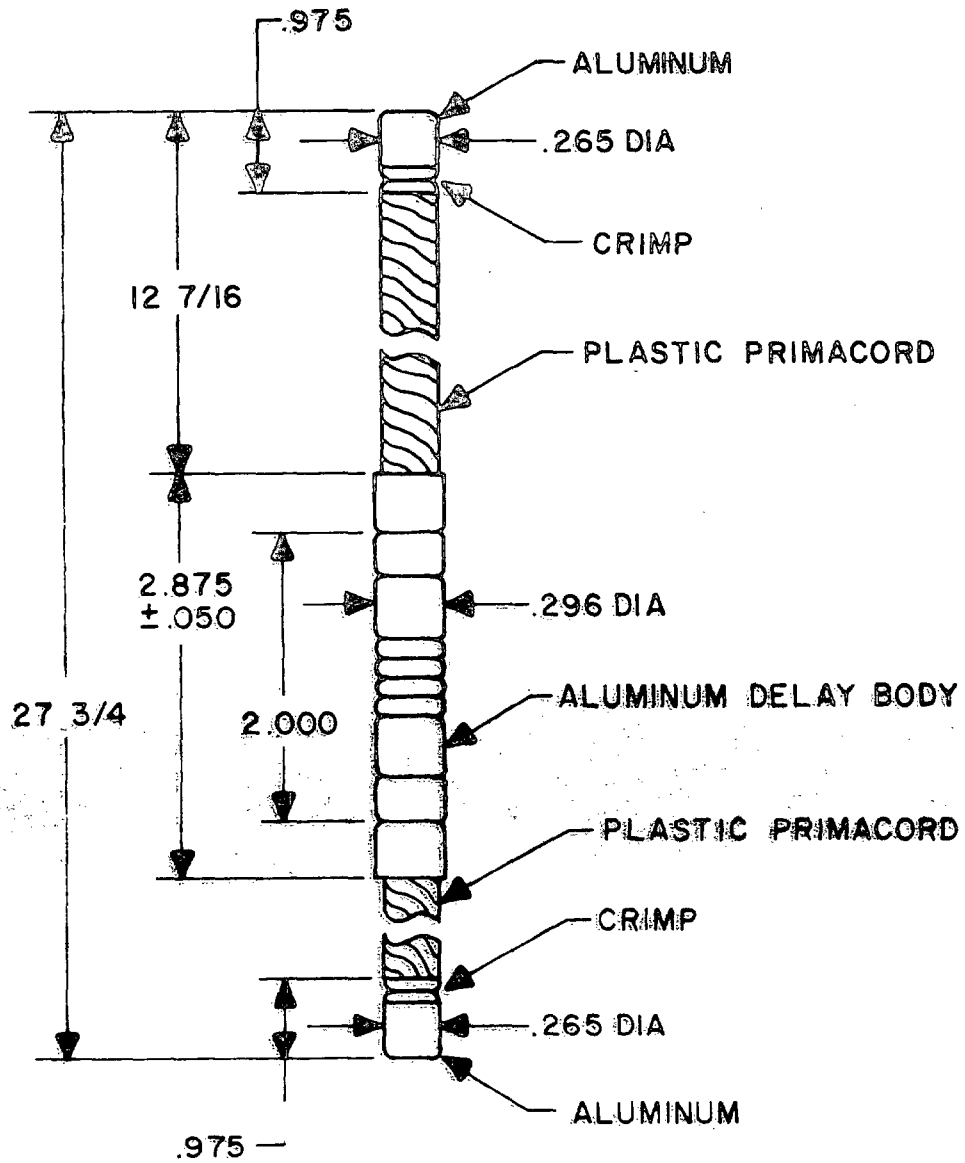
6.5 Supersession data. This specification supersedes MIL-D-23295(OS) dated 14 May 1962 and proposed MIL-D-23295A(OS) dated 18 October 1976, which was never printed.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing Activity:
Navy - OS

(Project No. 1375-N124)

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TOLERANCES:
FRACTIONS ±1/2
DECIMALS ±.015

NOTE: ALL DIMENSIONS ARE IN INCHES

FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-D-23295B(OS) Delay Element, Detonating, MK 19 MOD 0

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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